

n-th order linear differential equations

$$(1) \quad y^{(n)}(x) + p_{n-1}(x) \cdot y^{(n-1)} + \dots + p_0(x) \cdot y(x) = r(x)$$

homogeneous if $r(x) = 0$

Initial value problem:

$$y(x_0) = K_0, \dots, y^{(n-1)}(x_0) = K_{n-1} \Rightarrow y(x) = ?$$

Existence and unicity thm:

$p_i(x)$ - s continuous functions on I open interval $\Rightarrow$ solution exists, and is unique!

$$(1^*) \quad y^{(n)}(x) + p_{n-1}(x) \cdot y^{(n-1)} + \dots + p_0(x) \cdot y(x) = 0$$

- linear combination of some solutions is also a solution

- solutions $y_1, \dots, y_k$ are linearly dependent if $\exists d_i \neq 0$ such that

$$d_1 y_1(x) + \dots + d_k y_k(x) = 0 \quad \text{on } I$$

statement

$y_1, \dots, y_k$ are linearly dependent
 $\Leftrightarrow \exists x_0 \in I$ such that

$$W(x_0) = \begin{vmatrix} y_1(x_0) & y_2(x_0) & \dots & y_k(x_0) \\ y_1'(x_0) & & & \\ \vdots & & & \\ y_1^{(k-1)}(x_0) & \dots & \dots & \end{vmatrix} = 0$$

Example:

e.g. $e^{\lambda_1 x}$ and $e^{\lambda_2 x}$

$$\Rightarrow M = \begin{bmatrix} e^{\lambda_1 x} & e^{\lambda_2 x} \\ \lambda_1 e^{\lambda_1 x} & \lambda_2 e^{\lambda_2 x} \end{bmatrix} \Rightarrow$$

$$W = e^{(\lambda_1 + \lambda_2)x} \begin{vmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{vmatrix} = (\lambda_2 - \lambda_1) \cdot e^{(\lambda_1 + \lambda_2)x}$$

$\Rightarrow$ independent iff $\lambda_1 \neq \lambda_2$

Proof: (supplementary)

($\Rightarrow$) is easy:

$$\lambda_1 y_1 + \dots + \lambda_n y_n = 0$$

$\Downarrow$

$$\lambda_1 y_1' + \dots + \lambda_n y_n' = 0$$

$\Rightarrow$

$\vdots$

$$\lambda_1 y_1^{(n-1)} + \dots + \lambda_n y_n^{(n-1)} = 0$$

$$\underbrace{\begin{bmatrix} y_1 & y_2 & \dots \\ y_1' & & \\ \vdots & & \\ y_1^{(n-1)} & \dots \end{bmatrix}}_M \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix} = 0$$

has a nontrivial solution if M is not invertible, i.e. $\det(M) = W = 0$ for some x

$y_1(x), \dots, y_n(x)$ independent solutions of $(1)^*$ $\Rightarrow$ form a basis:

- any solution of $(1)^*$ can be expressed as a linear combination of these

- $c_1 y_1(x) + \dots + c_n y_n(x)$ forms the general solution of $(1)^*$

example:

$$y^{(4)} - 5y'' + 4y = 0$$

look for solution $y = e^{\lambda \cdot x} \Rightarrow$

$$\lambda^4 - 5\lambda^2 + 4 = (\lambda^2 - 1)(\lambda^2 - 4) = 0$$

$$\Rightarrow \lambda = \pm 1 \quad \lambda = \pm 2$$

general solution: $c_1 e^x + c_2 e^{-x} + c_3 e^{2x} + c_4 e^{-2x}$

initial value problem (tutorial!)

$$y(0) = 1; \quad y'(0) = y''(0) = y'''(0) = 0$$

$$\Rightarrow c_1 + c_2 + c_3 + c_4 = 1$$

$$c_1 - c_2 + 2(c_3 - c_4) = 0$$

$$c_1 + c_2 + 4(c_3 + c_4) = 0$$

$$c_1 - c_2 + 8(c_3 - c_4) = 0$$

$$\Rightarrow c_1 = c_2 = c, \quad c_3 = c_4 = -\frac{c}{4}$$

$$2c - \frac{1}{2}c = 1 \Rightarrow c = \frac{2}{3}$$

$$\Rightarrow y(x) = \frac{4}{3} \cosh(x) - \frac{1}{3} \cosh(2x)$$

Homogeneous linear ODE's with constant coefficients

$$y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_0 y = 0$$

Ansatz: $y = e^{\lambda \cdot x}$

$\Rightarrow$ solution if

$$(2) \quad \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_0 = 0$$

• cases:

(1) $\lambda_1, \dots, \lambda_n$ are all different $\Rightarrow$

$e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ are linearly independent:

$$W = e^{(\lambda_1 + \dots + \lambda_n) \cdot x} \begin{vmatrix} 1 & & 1 \\ \lambda_1 & & \\ \vdots & & \\ \lambda_1^{n-1} & \dots & \lambda_n^{n-1} \end{vmatrix} \propto \prod_{i < j} (\lambda_i - \lambda_j)$$

(2) $\lambda = \lambda_1$ root of order $m_1 \Rightarrow$

$e^{\lambda_1 x}, x e^{\lambda_1 x}, \dots, x^{m_1-1} e^{\lambda_1 x}$ are

linearly independent solutions!

Proof: (Supplemental)

Use the notion of differential operators

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mappings in the space
of functions

$f: X \mapsto f(x)$ differentiable

$g \equiv D[f]$ or Df is its differential

$$D: f \mapsto g \equiv f'$$

similarly, if f is n -times differentiable

then

$$f \mapsto L[f] \equiv f^{(n)} + p_{n-1} f^{(n-1)} + \dots + p_0 f$$

is a map

consider case with constant coeffs

$$L[f] \equiv f^{(n)} + a_{n-1} f^{(n-1)} + \dots + a_0 f$$

then

$$L[e^{\lambda \cdot x}] = \underbrace{(\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_0)}_{h(\lambda)} e^{\lambda \cdot x}$$

assume λ_1 has multiplicity w_1 then

$$L[e^{\lambda_1 \cdot x}] = (1 - \lambda_1)^{w_1} h_1(\lambda) \cdot e^{\lambda_1 \cdot x}$$

obviously $L[e^{\lambda_1 \cdot x}] = 0 \sim e^{\lambda_1 \cdot x}$ is a
solution

$$\frac{\partial}{\partial \lambda} L[e^{\lambda \cdot x}] = L[x \cdot e^{\lambda \cdot x}]$$

$$= m_1 (1 - \lambda_1)^{m_1-1} h_1(\lambda) e^{\lambda \cdot x} + (\lambda - \lambda_1)^{m_1} \frac{d}{d\lambda} (h_1(\lambda) e^{\lambda \cdot x})$$

now set $\lambda \rightarrow \lambda_1 \Rightarrow$ both vanish!

$$\Rightarrow L[x \cdot e^{\lambda_1 \cdot x}] = 0$$

can continue $m_1 - 1$ times

$\Rightarrow e^{\lambda_1 \cdot x}, x \cdot e^{\lambda_1 \cdot x}, \dots, x^{m_1-1} e^{\lambda_1 \cdot x}$
are all solutions of the hom.
equation

□

Solution of inhomogeneous equation:

$$M = \begin{bmatrix} y_1 & \dots & y_n \\ y_1' & & \vdots \\ \vdots & & \\ y_1^{(n-1)} & & y_n^{(n-1)} \end{bmatrix} \Rightarrow w_k \equiv \begin{vmatrix} y_1 & \dots & 0 & y_{k+1} \\ y_1' & & 0 & y_{k+1}' \\ \vdots & & \vdots & \vdots \\ y_1^{(n-1)} & & 1 & y_{k+1}^{(n-1)} \end{vmatrix}$$

$\uparrow$
 k

$$\Rightarrow y_p(x) = y_1 \int \frac{w_1(x)}{w(x)} r(x) dx + \dots + y_n \int \frac{w_n}{w} r dx$$

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Elastic beam

$$E \cdot I \cdot y^{(4)}(x) = f(x)$$

$\nwarrow$ load / length
 $\nearrow$ Young modulus

Inhomogeneous equation

$$y'' + p(x)y'(x) + q(x)y(x) = r(x)$$

statement: $y_1(x)$ and $y_2(x)$ are independent solutions of homogeneous equation $\Rightarrow$ closed form solution exists:

$$y_p(x) = -y_1 \int \frac{y_2 r}{w} dx + y_2 \int \frac{y_1 r}{w} dx$$

example:

$$y'' + y = x$$

$$y_1 = \cos(x) \quad , \quad y_2 = \sin(x)$$

$$w = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix} = 1$$

$$y_p(x) = -\cos(x) \cdot \int \sin(x) \cdot x \, dx + \sin(x) \int \cos(x) \cdot x \, dx$$

$$\begin{aligned} &= -\cos(x) \cdot x + \int \cos(x) \\ &= \sin(x) - x \cos(x) \end{aligned}$$

$$\begin{aligned} &\sin(x) \cdot x - \int \sin(x) \\ &= \cos(x) + x \cdot \sin(x) \end{aligned}$$

$$\begin{aligned} y_p(x) &= x \cdot \cos^2(x) - \cos(x) \cdot \sin(x) + \sin(x) \cdot \cos(x) + \\ &\quad x \cdot \sin^2(x) = x \quad ! \quad \checkmark \end{aligned}$$